```
# Amazon Alexa Customer Review Sentiment Analysis
# Objective: Classify reviews as positive or negative using Random Forest

import tensorflow as tf
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from tensorflow import keras
```

Mount Google Drive and Load Dataset

```
import os
os.listdir('/content/drive/MyDrive/')
```

Show hidden output

```
# Mount Google Drive
from google.colab import drive
drive.mount('/content/drive')

# Load the dataset
alexa_df = pd.read_csv('/content/drive/MyDrive/amazon_alexa.tsv', sep='\t')
alexa_df.head()
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

	rating	date	variation	verified_reviews	feedback
0	5	31-Jul-18	Charcoal Fabric	Love my Echo!	1
1	5	31-Jul-18	Charcoal Fabric	Loved it!	1
2	4	31-Jul-18	Walnut Finish	Sometimes while playing a game, you can answer...	1
3	5	31-Jul-18	Charcoal Fabric	I have had a lot of fun with this thing. My 4 ...	1
4	5	31-Jul-18	Charcoal Fabric	Music	1

Preview the reviews and feedback

```
alexa_df['verified_reviews']
```

	verified_reviews
0	Love my Echo!
1	Loved it!
2	Sometimes while playing a game, you can answer...
3	I have had a lot of fun with this thing. My 4 ...
4	Music
...	...
3145	Perfect for kids, adults and everyone in betwe...
3146	Listening to music, searching locations, check...
3147	I do love these things, i have them running my...
3148	Only complaint I have is that the sound qualit...
3149	Good

3150 rows × 1 columns

dtype: object

Separate positive and negative reviews

```
positive = alexa_df[alexa_df['feedback'] == 1]
```

```
positive = alexa_df[alexa_df['feedback'] == 1]
negative = alexa_df[alexa_df['feedback'] == 0]

positive
```

	rating	date	variation	verified_reviews	feedback
0	5	31-Jul-18	Charcoal Fabric	Love my Echo!	1
1	5	31-Jul-18	Charcoal Fabric	Loved it!	1
2	4	31-Jul-18	Walnut Finish	Sometimes while playing a game, you can answer...	1
3	5	31-Jul-18	Charcoal Fabric	I have had a lot of fun with this thing. My 4 ...	1
4	5	31-Jul-18	Charcoal Fabric	Music	1
...	...	...	...	...	...
3145	5	30-Jul-18	Black Dot	Perfect for kids, adults and everyone in betwe...	1
3146	5	30-Jul-18	Black Dot	Listening to music, searching locations, check...	1
3147	5	30-Jul-18	Black Dot	I do love these things, i have them running my...	1
3148	5	30-Jul-18	White Dot	Only complaint I have is that the sound qualit...	1
3149	4	29-Jul-18	Black Dot	Good	1

2893 rows × 5 columns

```
negative
```

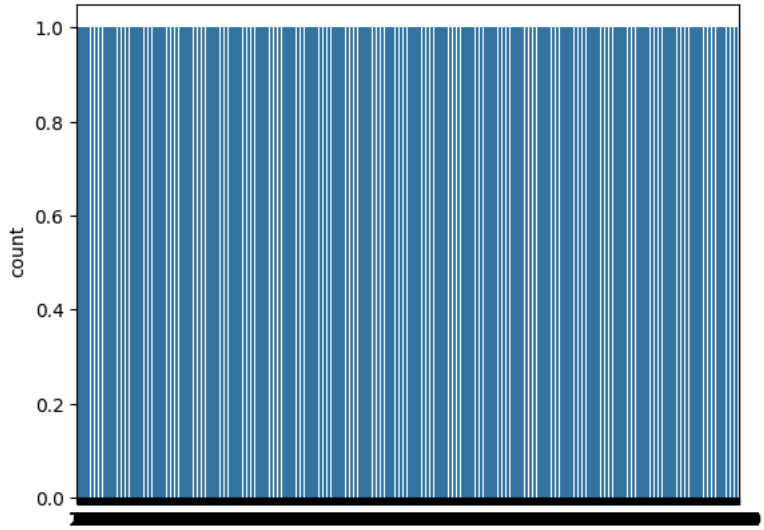
	rating	date	variation	verified_reviews	feedback
46	2	30-Jul-18	Charcoal Fabric	It's like Siri, in fact, Siri answers more acc...	0
111	2	30-Jul-18	Charcoal Fabric	Sound is terrible if u want good music too get...	0
141	1	30-Jul-18	Charcoal Fabric	Not much features.	0
162	1	30-Jul-18	Sandstone Fabric	Stopped working after 2 weeks ,didn't follow c...	0
176	2	30-Jul-18	Heather Gray Fabric	Sad joke. Worthless.	0
...	...	...	...	...	...
3047	1	30-Jul-18	Black Dot	Echo Dot responds to us when we aren't even ta...	0
3048	1	30-Jul-18	White Dot	NOT CONNECTED TO MY PHONE PLAYLIST :(	0
3067	2	30-Jul-18	Black Dot	The only negative we have on this product is t...	0
3091	1	30-Jul-18	Black Dot	I didn’t order it	0
3096	1	30-Jul-18	White Dot	The product sounded the same as the emoji spea...	0

257 rows × 5 columns

Data Visualization and Clean Up

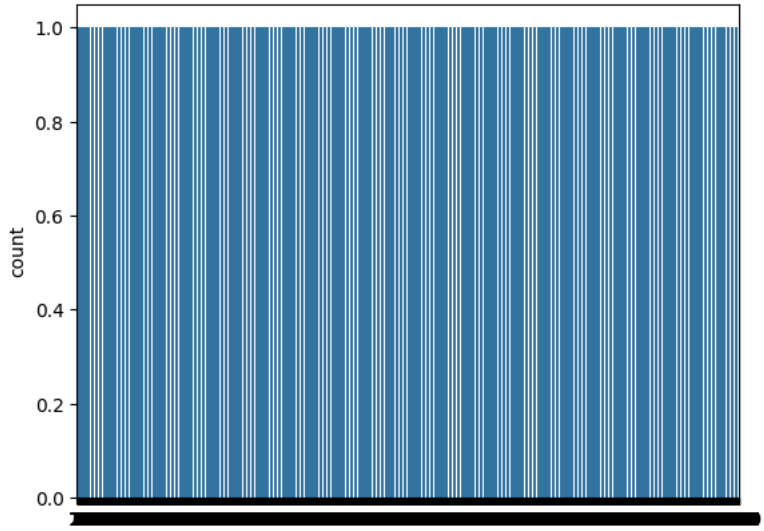
```
# Visualize the distribution of feedback
sns.countplot(alexa_df['feedback'], label='Count')
```

```
<Axes: ylabel='count'>
```



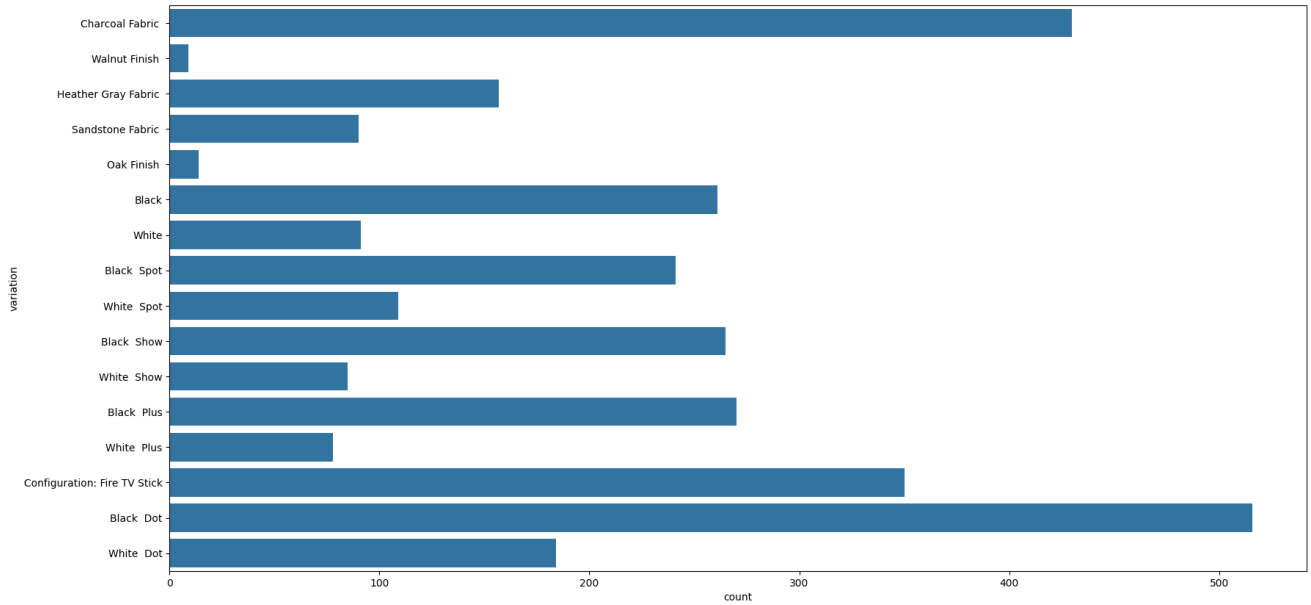

```
# Visualize the distribution of ratings
sns.countplot(alexa_df['rating'], label='Count')
```

```
<Axes: ylabel='count'>
```




```
# Visualize the distribution of variation
plt.figure(figsize=(20, 10))
sns.countplot(alexa_df['variation'], label='Count')
```

```
<Axes: xlabel='count', ylabel='variation'>
```

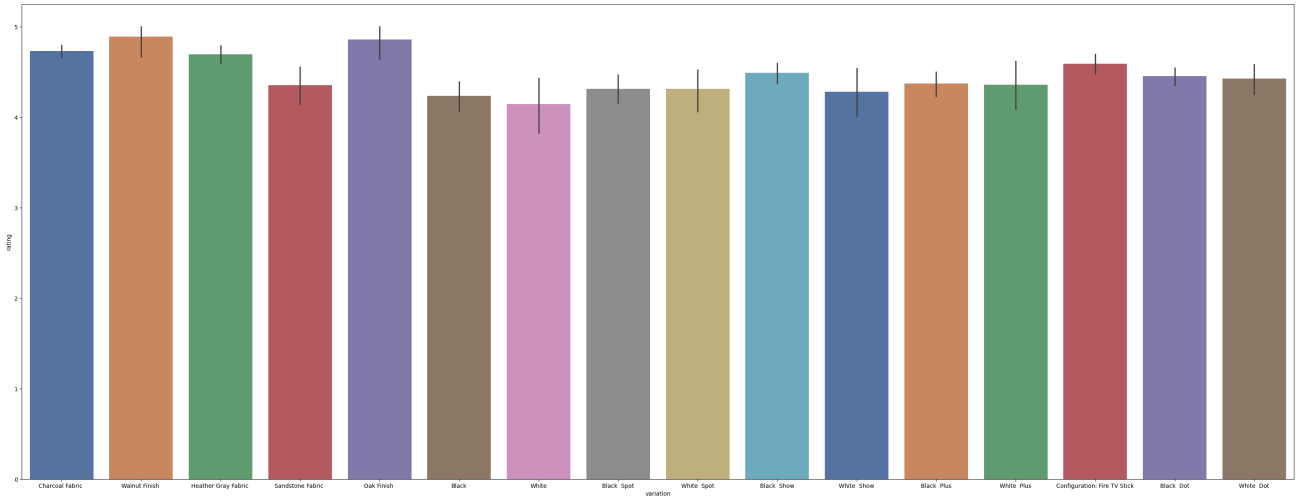



```
# Visualize the relationship between variations and ratings
plt.figure(figsize=(40, 15))
sns.barplot(x='variation', y='rating', data=alexa_df, palette='deep')
```

```
/tmp/ipykernel_45272/3600321335.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and se

  sns.barplot(x='variation', y='rating', data=alexa_df, palette='deep')
<Axes: xlabel='variation', ylabel='rating'>
```



Drop irrelevant columns

```
alexa_df = alexa_df.drop(['date', 'rating'], axis=1)
alexa_df
```

	variation	verified_reviews	feedback
0	Charcoal Fabric	Love my Echo!	1
1	Charcoal Fabric	Loved it!	1
2	Walnut Finish	Sometimes while playing a game, you can answer...	1
3	Charcoal Fabric	I have had a lot of fun with this thing. My 4 ...	1
4	Charcoal Fabric	Music	1
...	...	...	...
3145	Black Dot	Perfect for kids, adults and everyone in betwe...	1
3146	Black Dot	Listening to music, searching locations, check...	1
3147	Black Dot	I do love these things, i have them running my...	1
3148	White Dot	Only complaint I have is that the sound qualit...	1
3149	Black Dot	Good	1

3150 rows × 3 columns

One-Hot Encoding

```
# Perform one-hot encoding on the 'variation' column
variation_dummies = pd.get_dummies(alexa_df['variation'])
alexa_df = pd.concat([alexa_df, variation_dummies], axis=1)
alexa_df = alexa_df.drop(['variation'], axis=1)
alexa_df
```

	verified_reviews	feedback	Black	Black Dot	Black Plus	Black Show	Black Spot	Charcoal Fabric	Configuration: Fire TV Stick	Heather Gray Fabric	Oak Finish	Sandstone Fabric	Walnut Finish
0	Love my Echo!	1	False	False	False	False	False	True	False	False	False	False	False
1	Loved it!	1	False	False	False	False	False	True	False	False	False	False	False
2	Sometimes while playing a game, you can answer...	1	False	False	False	False	False	False	False	False	False	False	True
3	I have had a lot of fun with this thing. My 4 ...	1	False	False	False	False	False	True	False	False	False	False	False
4	Music	1	False	False	False	False	False	True	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...
3145	Perfect for kids, adults and everyone in betwe...	1	False	True	False	False	False	False	False	False	False	False	False

Tokenization using CountVectorizer

```
# Tokenize the 'verified_reviews' column using CountVectorizer
from sklearn.feature_extraction.text import CountVectorizer

# Fill any NaN values in verified_reviews with empty string
alexa_df['verified_reviews'] = alexa_df['verified_reviews'].fillna('')

vectorizer = CountVectorizer()
alexa_reviews = vectorizer.fit_transform(alexa_df['verified_reviews'])

# Combine tokenized reviews with the main dataframe
alexa_reviews_df = pd.DataFrame(alexa_reviews.toarray())
alexa_df = alexa_df.drop(['verified_reviews'], axis=1)
alexa_df = pd.concat([alexa_df, alexa_reviews_df], axis=1)
alexa_df
```

	feedback	Black	Black Dot	Black Plus	Black Show	Black Spot	Charcoal Fabric	Configuration: Fire TV Stick	Heather Gray Fabric	Oak Finish	...	4034	4035	4036	4037	4038	4
0	1	False	False	False	False	False	True	False	False	False	...	0	0	0	0	0	
1	1	False	False	False	False	False	True	False	False	False	...	0	0	0	0	0	
2	1	False	False	False	False	False	False	False	False	False	...	0	0	0	0	0	
3	1	False	False	False	False	False	True	False	False	False	...	0	1	0	0	0	
4	1	False	False	False	False	False	True	False	False	False	...	0	0	0	0	0	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
3145	1	False	True	False	False	False	False	False	False	False	...	0	0	0	0	0	
3146	1	False	True	False	False	False	False	False	False	False	...	0	0	0	0	0	
3147	1	False	True	False	False	False	False	False	False	False	...	0	0	0	0	0	
3148	1	False	False	False	False	False	False	False	False	False	...	0	0	0	0	0	
3149	1	False	True	False	False	False	False	False	False	False	...	0	0	0	0	0	

Prepare data for modeling

```
# Split the data into features (X) and target (y)
X = alexa_df.drop(['feedback'], axis=1)
Y = alexa_df['feedback']

print(X.shape, Y.shape)
```

```
(3150, 4060) (3150,)
```

Train and Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, Y,
                                                    test_size=0.20,
                                                    random_state=5)

print(X.shape, Y.shape)
print(X_train.shape, y_train.shape, X_test.shape)
```

```
(3150, 4060) (3150,)
(2520, 4060) (2520,) (630, 4060)
```

Build the Model

```
# Model
model = tf.keras.models.Sequential()
model.add(tf.keras.layers.Dense(units=400, activation='relu',
                                input_shape=(X_train.shape[1], )))
model.add(tf.keras.layers.Dense(units=200, activation='relu'))
model.add(tf.keras.layers.Dense(units=1, activation='sigmoid'))

model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential"
```

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 400)	1,624,400
dense_1 (Dense)	(None, 200)	80,200
dense_2 (Dense)	(None, 1)	201

```
Total params: 1,704,801 (6.50 MB)
Trainable params: 1,704,801 (6.50 MB)
Non-trainable params: 0 (0.00 B)
```

Compile the Model

```
model.compile(optimizer='Adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])
```

Train the Model

```
# Convert boolean columns to numerical
X_train = X_train.astype(float)
y_train = y_train.astype(float)

# Now fit the model
epochs_hist = model.fit(X_train, y_train, epochs=10)
```

```
Epoch 1/10
79/79 ━━━━━━━━━━━━━━━━━━━━ 3s 21ms/step - accuracy: 0.9147 - loss: 0.2882
Epoch 2/10
79/79 ━━━━━━━━━━━━━━━━━━━━ 2s 27ms/step - accuracy: 0.9635 - loss: 0.0905
Epoch 3/10
79/79 ━━━━━━━━━━━━━━━━━━━━ 2s 29ms/step - accuracy: 0.9889 - loss: 0.0404
Epoch 4/10
79/79 ━━━━━━━━━━━━━━━━━━━━ 2s 22ms/step - accuracy: 0.9937 - loss: 0.0206
Epoch 5/10
79/79 ━━━━━━━━━━━━━━━━━━━━ 2s 20ms/step - accuracy: 0.9940 - loss: 0.0172
Epoch 6/10
79/79 ━━━━━━━━━━━━━━━━━━━━ 2s 20ms/step - accuracy: 0.9933 - loss: 0.0166
Epoch 7/10
79/79 ━━━━━━━━━━━━━━━━━━━━ 3s 20ms/step - accuracy: 0.9940 - loss: 0.0159
Epoch 8/10
79/79 ━━━━━━━━━━━━━━━━━━━━ 2s 20ms/step - accuracy: 0.9940 - loss: 0.0161
Epoch 9/10
79/79 ━━━━━━━━━━━━━━━━━━━━ 3s 29ms/step - accuracy: 0.9948 - loss: 0.0154
Epoch 10/10
79/79 ━━━━━━━━━━━━━━━━━━━━ 2s 28ms/step - accuracy: 0.9940 - loss: 0.0147
```

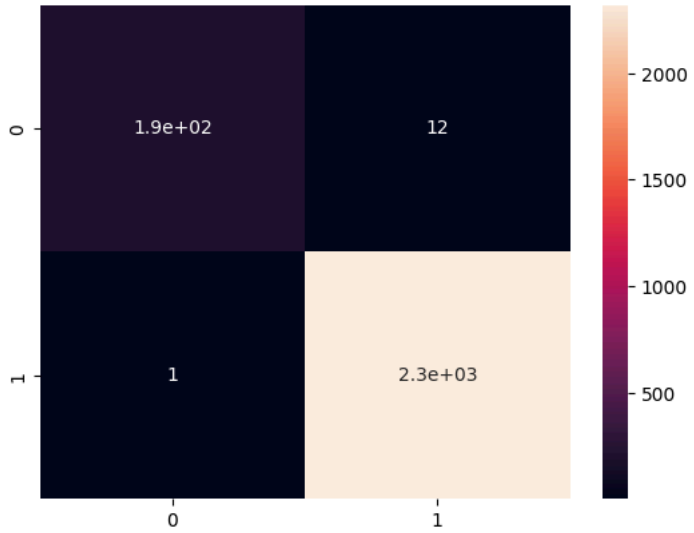
Evaluate Model - Training set

```
from sklearn.metrics import classification_report, confusion_matrix

y_pred_train = model.predict(X_train)
y_pred_train = (y_pred_train > 0.5)

cm = confusion_matrix(y_train, y_pred_train)
sns.heatmap(cm, annot=True)
```

```
79/79 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step
<Axes: >
```



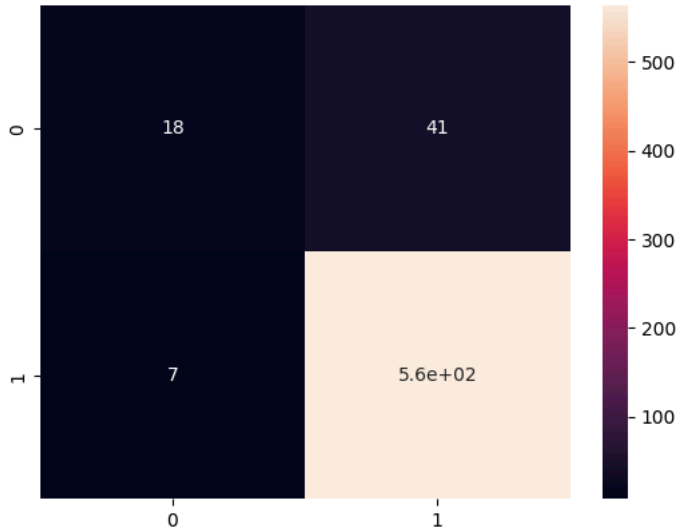
Evaluate Model - Test set

```
# Convert boolean columns in X_test to numerical
X_test = X_test.astype(float)

# Now predict
y_pred_test = model.predict(X_test)
y_pred_test = (y_pred_test > 0.5)

cm = confusion_matrix(y_test, y_pred_test)
sns.heatmap(cm, annot=True)
```

```
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step
<Axes: >
```



Classification Report

```
print(classification_report(y_test, y_pred_test))
```

```
              precision    recall  f1-score   support

           0       0.72      0.31      0.43        59
           1       0.93      0.99      0.96       571

    accuracy                           0.92       630
   macro avg       0.83      0.65      0.69       630
```

```
weighted avg       0.91      0.92      0.91       630
```

Training History Keys

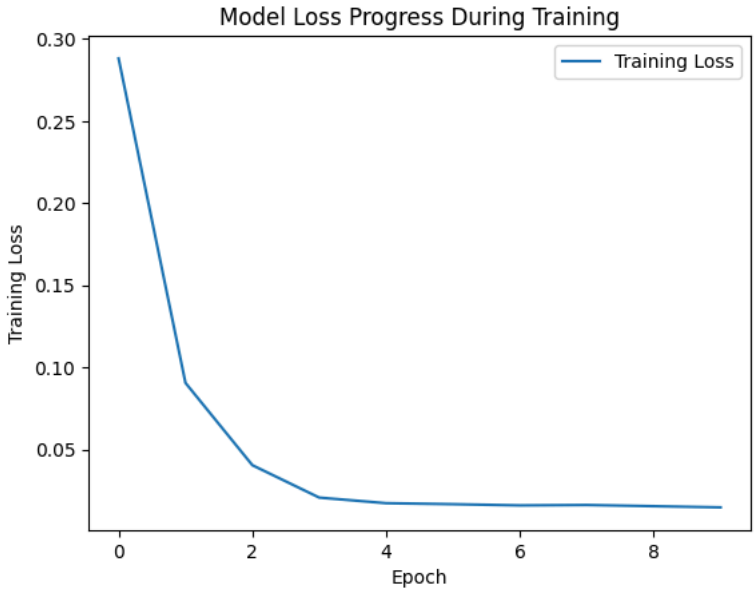
```
epochs_hist.history.keys()
```

```
dict_keys(['accuracy', 'loss'])
```

Plot Model Loss Progress

```
plt.plot(epochs_hist.history['loss'])
plt.title('Model Loss Progress During Training')
plt.xlabel('Epoch')
plt.ylabel('Training Loss')
plt.legend(['Training Loss'])
```

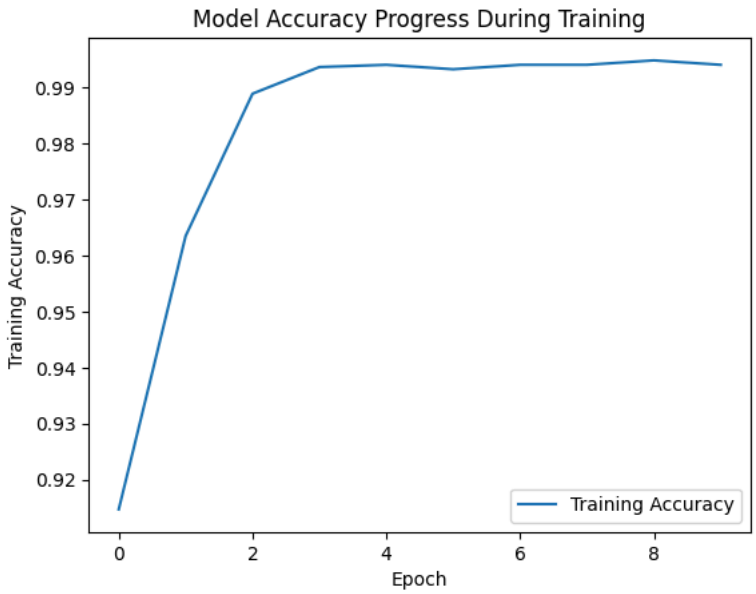
```
<matplotlib.legend.Legend at 0x79e0d179cf50>
```



Plot Model Accuracy Progress

```
plt.plot(epochs_hist.history['accuracy'])
plt.title('Model Accuracy Progress During Training')
plt.xlabel('Epoch')
plt.ylabel('Training Accuracy')
plt.legend(['Training Accuracy'])
```

```
<matplotlib.legend.Legend at 0x79e0d18621b0>
```



Compare training and testing accuracy

```
# Task 8: Analyze Model Performance
train_accuracy = model.score(X_train, y_train) \
    if hasattr(model, 'score') else \
    model.evaluate(X_train, y_train.astype(float))[1]

test_accuracy = model.evaluate(X_test, y_test.astype(float))[1]

print(f"Training Accuracy: {train_accuracy}")
print(f"Testing Accuracy: {test_accuracy}")
```

```
79/79 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.9948 - loss: 0.0128
20/20 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9238 - loss: 0.5369
Training Accuracy: 0.9948412775993347
Testing Accuracy: 0.9238095283508301
```

Feature Importance using Random Forest

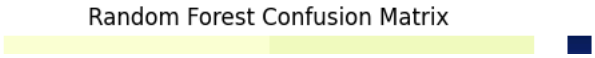
```
from sklearn.ensemble import RandomForestClassifier

# Define and train a Random Forest Classifier
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)

# Convert all column names to strings
X_train.columns = X_train.columns.astype(str)
X_test.columns = X_test.columns.astype(str)


rf_model.fit(X_train, y_train)

# Generate predictions
rf_pred = rf_model.predict(X_test)

# Confusion matrix
rf_cm = confusion_matrix(y_test, rf_pred)
sns.heatmap(rf_cm, annot=True, cmap='YlGnBu', fmt='g')
plt.title('Random Forest Confusion Matrix')
plt.ylabel('Actual label')
plt.xlabel('Predicted label')
```

```
Text(0.5, 23.52222222222222, 'Predicted label')
```